

Norm Vector Space

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26 Oct 2024

- **Vector Spaces**
- **Normed Vector Spaces**
- **P-Norm**
- **Linear Functionals and Continuity**
- **Banach Spaces**

- A vector is a *mathematical object* with 2 features:
 - **Direction and Magnitude**
- **Why do we need vectors instead of just points?**
 - Points (also a *mathematical object*) only tell us the **position** in space.
 - Vectors tell us : Where to face and **how much to move** to reach the position.

Def.1

A **vector space** is a set over a field F with two operations:

- Addition
- Scalar multiplication

The following must hold:

- $\forall x, y \in V : x + y \in V$
- $\forall x \in V, c \in F : c \cdot x \in V$
- 8 more properties (e.g., $x + y = y + x$)

Normed Vector Space

Def.2

A **norm** on a vector space X is a real-valued function, denoted $\|x\|$, that satisfies these properties for all $x, y \in X$, and $a \in \mathbb{R}$:

- ① $\|x\| \geq 0$
- ② $\|x\| = 0 \iff x = 0$
- ③ $\|ax\| = |a|\|x\|$
- ④ $\|x + y\| \leq \|x\| + \|y\|$

Def.3

A **normed vector space** is a vector space equipped with a norm.

Example 1: Normed Vector Space

- Is $\mathbb{R}$ with the norm as the absolute value a Normed Vector Space?

Let $V = \mathbb{R}$, $F = \mathbb{R}$, and $\|x\| = |x|$.

Is $\mathbb{R}$ a Vector Space?

- 1 $\mathbb{R}$ satisfies Definition 1.

Is $|x|$ a norm?

- 1 $\|x\| = |x| \geq 0$
- 2 $\|x\| = 0 \iff |x| = 0 \iff x = 0$
- 3 $\|ax\| = |a||x| = |a|\|x\|$
- 4 $\|x + y\| = |x + y| \leq |x| + |y| = \|x\| + \|y\|$

Example 2: Normed Vector Space

Let $V = \mathbb{R}^2$, $\forall v = (x, y) \in \mathbb{R}^2$, the norm $\|v\|$ is defined as $\sqrt{x^2 + y^2}$.

- ① $\|v\| = \sqrt{x^2 + y^2} \geq 0$
- ② $\|v\| = 0 \iff \sqrt{x^2 + y^2} = 0 \iff x = 0 \text{ and } y = 0$
- ③ $\|av\| = \sqrt{(ax)^2 + (ay)^2} = \sqrt{a^2x^2 + a^2y^2} = \sqrt{a^2(x^2 + y^2)} = |a|\sqrt{x^2 + y^2} = |a|\|v\|$
- ④ $\|x + y\| \leq \dots \leq \|x\| + \|y\|$

Example 3: $\mathbb{R}^n$ with Euclidean Norm

$\mathbb{R}^n$ with the Euclidean norm for any dimension n is a Normed Vector Space.

$$\|v\| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad \text{for } v = (x_1, \dots, x_n)$$

- **P-norm:** $\|x\|_p = (|x_1|^p + |x_2|^p + \cdots + |x_n|^p)^{\frac{1}{p}}$
- P-norm is a generalized way of measuring the length of a vector in a vector space.
- Set different p 's:
 - $p = 1$: $\|x\|_1 = |x_1| + |x_2| + \cdots + |x_n|$ (Taxicab norm)
 - $p = 2$: $\|x\|_2 = \sqrt{|x_1|^2 + |x_2|^2 + \cdots + |x_n|^2}$ (Euclidean norm)
 - $p \rightarrow \infty$: $\|x\|_\infty = \max(|x_1|, |x_2|, \dots, |x_n|)$ (Max norm or Sup norm)
- p controls how vector components contribute to the overall length.

Def.4

A **linear functional** f is a mapping $f : X \rightarrow \mathbb{R}$ or $f : X \rightarrow \mathbb{C}$ such that $\forall x, y \in X$:

- ① $f(cx) = cf(x)$
- ② $f(x + y) = f(x) + f(y)$

Def.5

A function $T : V \rightarrow W$ is a **linear transformation** if $\forall u, v \in V$ and $c \in F$:

- ① $T(cu) = cT(u)$
- ② $T(u + v) = T(u) + T(v)$

Example 1: Linear Functionals

- **E.g.:** f is linear means f grows at a constant rate.
- $c = f(x, y) = 3x + 5y$
- $d = g(x, y) = 3x^2 + 5y$

Continuity of Linear Functionals

Thm.1: Continuity of Linear Functionals

Let f be linear functional on X where X is a norm linear space
 f is continuous $\iff \sup_{\|x\|\leq 1} |f(x)| < \infty$

Continuity of Linear Functionals (Proof.)

Suppose $\sup_{\|x\| \leq 1} |f(x)| = M < \infty$.

We want to show f is continuous at any point x_0 .

That is, for any $\epsilon > 0$, there exists $\delta > 0$ such that

$$|f(x) - f(x_0)| < \epsilon \quad \text{when} \quad \|x - x_0\| < \delta.$$

Let

$$y = \frac{x - x_0}{\|x - x_0\|}.$$

Then:

$$\|y\| = \frac{\|x - x_0\|}{\|\|x - x_0\|\|} = \frac{\|x - x_0\|}{\|x - x_0\|} = 1.$$

Now we see $\|y\| = 1$, which means y is a unit vector.

Continuity of Linear Functionals (Proof.)

- By assumption: $\sup_{\|x\| \leq 1} |f(x)| = M < \infty$, so $|f(y)| \leq M$. (1)

Let:

$$y = \frac{x - x_0}{\|x - x_0\|}.$$

Apply f to both sides:

$$f(y) = f\left(\frac{x - x_0}{\|x - x_0\|}\right) \longrightarrow \frac{1}{\|x - x_0\|} f(x - x_0) \longrightarrow \frac{1}{\|x - x_0\|} (f(x) - f(x_0)).$$

Now apply absolute values:

$$|f(y)| = \left| \frac{1}{\|x - x_0\|} (f(x) - f(x_0)) \right| \leq M \quad (\text{using (1)}).$$

Finally, multiply both sides by $\|x - x_0\|$:

$$|f(x) - f(x_0)| \leq M \cdot \|x - x_0\|.$$

Thus:

$$|f(x) - f(x_0)| \leq M \cdot \|x - x_0\| \leq M \cdot \delta < \epsilon.$$

Neighbourhoods vs. Sequences

- **Question:** Why use neighbourhood instead of sequence for continuity?
 - Neighbourhood captures *locality* better than a sequence.
 - Even in vector spaces, such as applications like wind flow, the concept of "closeness" is better represented by neighbourhoods than by sequences.
- **Neighbourhood:**
 - f is continuous at $f(\vec{p})$ if
For each neighbourhood W of $f(\vec{p})$, there exists a neighbourhood V of $\vec{p}$ such that $f(V) \subseteq W$.
- **Sequence:**
 - f is continuous at x_0 if
for every sequence $\{x_n\}$ converging to x_0 , the sequence $\{f(x_n)\}$ converges to $f(x_0)$.

Neighbourhoods vs. Sequences

- **Neighbourhood:** Focuses on surrounding vectors of $\vec{p}$ for continuity.
- **Sequence:** Relies on individual points rather than the locality of vectors.

Def.6

A **Cauchy sequence** is a sequence whose terms get closer to each other as the sequence progresses.

Def.7

A space is **complete** if every Cauchy sequence within the space converges to a point that is also within that space.

Thm.2: Cauchy Sequence described with vector sequence

Let X be a normed linear space.

A sequence of vectors $\{x_n\}$ in X is a **Cauchy sequence** if and only if

$$\|x_n - x_m\| \rightarrow 0 \quad \text{as } m \rightarrow \infty \text{ and } n \rightarrow \infty.$$

Thm.3 X is a Complete Space

Let X be a normed linear space. X is a **complete space** if and only if every Cauchy sequence in X converges to a specific vector x that is also within X .

Def.8

A normed linear space that is complete is called a **Banach space**.

Example: Banach Space

$C[a, b]$ equipped with the supremum norm:

$$\|f\| = \sup_{x \in [a, b]} |f(x)|$$

is a **Banach Space**.

- **Meaning:** Every sequence of continuous functions on $[a, b]$ converges to a specific continuous function within $C[a, b]$.